

Research Proposal for Capstone and Future Publication

Title: Investigating the Impact of Self-Supervised Pretraining on CNN and Vision Transformer Encoders for Multi-Disease Chest X-Ray Classification Under Low-Label Conditions

1. Introduction

Recent advances in Artificial Intelligence have significantly improved automated medical image analysis. Deep learning models can detect various thoracic diseases from chest X-ray images, assisting radiologists in diagnosis and reducing workload. However, most supervised learning approaches require large quantities of expert-annotated medical images, which are expensive and time-consuming to obtain.

At the same time, hospitals possess millions of unlabeled chest X-ray images. Self-Supervised Learning (SSL) offers a promising solution by allowing models to learn useful visual representations from unlabeled data before being fine-tuned on smaller labeled datasets.

This research aims to investigate how different encoder architectures benefit from SSL pretraining and whether SSL can reduce the dependence on labeled medical data while maintaining strong disease detection performance.

2. Motivation

Most existing studies focus on:

- Binary classification (Normal vs Pneumonia)
- Single model architectures
- Fully labeled datasets

However, several important questions remain unanswered:

1. Do CNNs and Vision Transformers benefit equally from SSL?
2. Which architecture gains the most improvement from SSL pretraining?
3. Can SSL significantly reduce the number of labeled medical images required?
4. Does SSL improve performance equally across different diseases?

Answering these questions could contribute to the development of more efficient medical AI systems.

3. Research Direction 1:

CNNs vs Vision Transformers Under SSL Pretraining

Objective

To compare how different encoder architectures benefit from self-supervised pretraining.

CNN Encoders

- ResNet-50
- DenseNet-121

Transformer Encoders

- Vision Transformer (ViT)
- Swin Transformer

Research Question

Which encoder architecture benefits the most from self-supervised pretraining on chest X-ray images?

Hypothesis

Transformers may benefit more from SSL because they generally require larger amounts of training data compared to CNNs.

Experimental Setup

For each encoder:

1. Train from scratch
2. SSL pretrain on unlabeled chest X-rays
3. Fine-tune on disease labels
4. Compare performance

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Expected Contribution

This study will provide insights into which encoder architecture is most suitable for medical image representation learning.

4. Research Direction 2:

Multi-Disease Chest X-Ray Classification

Objective

Most previous work focuses on binary classification. Real clinical environments involve multiple diseases simultaneously.

Diseases to Detect

1. Pneumonia
2. Tuberculosis
3. Cardiomegaly
4. Pleural Effusion

Research Question

Does self-supervised learning improve performance equally across different thoracic diseases?

Motivation

Different diseases appear differently in chest X-rays.

For example:

- Pneumonia often appears as localized opacity.
- Cardiomegaly affects heart size.
- Pleural Effusion appears near lung boundaries.

SSL may learn representations that are more useful for some diseases than others.

Experimental Setup

For each disease:

1. Train supervised baseline
2. Train SSL-pretrained model
3. Compare performance gains

Expected Contribution

The research will identify disease categories that benefit most from SSL representations.

This information can guide future medical AI development.

5. Research Direction 3:

Low-Label Learning Analysis

Objective

To determine how many labeled chest X-rays are actually needed after SSL pretraining.

Motivation

Medical annotation is expensive because it requires expert radiologists.

If SSL can reduce labeling requirements, hospitals can deploy AI systems more efficiently.

Research Question

How much labeled data can be replaced by SSL pretraining?

Label Ratios

Experiments will be conducted using:

- 100% labels
- 50% labels
- 20% labels
- 10% labels
- 5% labels

Experimental Design

For each label ratio:

Supervised Baseline

Train model from scratch.

SSL Pipeline

1. SSL pretraining on unlabeled images
2. Fine-tuning on reduced labeled data

Key Comparison

Can:

SSL + 10% Labels

achieve similar performance to:

Supervised + 100% Labels?

Expected Contribution

This study could demonstrate significant reductions in labeling requirements while maintaining clinical performance.

8. Expected Outcomes

The research aims to answer:

1. Which encoder benefits most from SSL?
2. Are Transformers superior to CNNs after SSL pretraining?
3. Which diseases benefit most from SSL?
4. How many labels can SSL save?
5. Can SSL reduce annotation costs in medical imaging?

9. Publication Potential

This project combines:

- Self-Supervised Learning
- Vision Transformers
- CNN Architectures
- Multi-Disease Detection
- Low-Label Learning

These are active research areas in:

- Medical Image Analysis
- Computer Vision
- Representation Learning
- Healthcare AI

The proposed work has potential for publication in undergraduate research conferences and could be extended into a journal article with sufficient experimental validation.

Proposed Final Research Title

“Evaluating Self-Supervised Pretraining for CNN and Vision Transformer Encoders in Multi-Disease Chest X-Ray Classification Under Low-Label Conditions”